What "hold D/r fixed and let the octupole miss a little" means (CoE model vs Migkas+2020 $A(\ell)$)

Sedov–Taylor Center-of-Expansion model, analytic point-mass tidal form, sources on the vision shell ($d = r$), with Chris's rotation and 2ℓ transforms. Scripts: `figures/fixedDr_octmiss.py` (solve + figure; reuses the data, coefficients and solver of `termmatch_Dr234.py`) and `make_fixedDr_octmiss_tex.py` (writes this sheet from `fixedDr_octmiss_results.json`).

In one sentence. With D/r free, the four harmonic amplitudes ($m = 0..3$) fix the four unknowns exactly and the octupole forces $D/r \approx 1$. If we instead *choose* $D/r = 2, 3, 4$, we have one unknown fewer than equations; we match the mean, dipole and quadrupole, and the octupole amplitude becomes a *prediction* that comes out too small. That shortfall is the "miss".

Step 0. Chris's variables

r = vision-shell radius = 14.3 Gpc; D = our distance from the CoE; L = density scale length; M/M_0 = enclosed mass toward the CoE (spreadsheet); (ℓ_0, b_0) = longitude and tilt of the CoE axis; φ = rotation phase in $\mathrm{Amp}_1 \sin(\ell + \varphi)$; φ_2 = 2ℓ phase in $\mathrm{Amp}_2 \sin(2\ell + \varphi_2)$; Amp = harmonic amplitudes.

Step 1. Define parameters

$$\varepsilon \equiv r/D = 1/(D/r), \qquad M_{\rm crit}(<D) = \tfrac{4\pi}{3}\rho_c D^3 = \frac{H_0^2 D^3}{2G},$$
$$\kappa \equiv \frac{G\,M(<D)}{H_0^2 D^3} = \frac{M(<D)}{2M_{\rm crit}}, \qquad K \equiv \alpha\kappa \ (\alpha = O(1);\ \text{tables use } \alpha = 1),$$
$$N = \text{overall normalisation}, \qquad a \equiv \cos b_0, \qquad \mu = \cos b \cos b_0 \cos\psi + \sin b \sin b_0, \qquad \psi = \ell - \ell_0.$$

Model, then averaged over the Migkas 90° sectors ($|b| > 20°$, $\cos b$ weight; same coefficients as the v2 and term-matching sheets):

$$\frac{A}{A_{\rm all}} = N\big[1 + K(3\mu^2 - 1) + K\varepsilon(\tfrac{9}{2}\mu - \tfrac{15}{2}\mu^3)\big] \xrightarrow{\text{sector avg}} N(1 + Kq_0) + NK\varepsilon\, o_1 \cos\psi + NKq_2 \cos 2\psi + NK\varepsilon\, o_3 \cos 3\psi,$$
$$q_0 = 0.4590 - 0.6885a^2, \quad q_2 = 0.49052a^2, \quad o_1 = 0.90032\,a(-3.1958 + 3.9948a^2), \quad o_3 = -0.22806a^3.$$

L does not appear: the point-mass form depends on the mass only through K, so L (and M/M_0 separately from L) is not determined here.

Step 2. Data harmonics and their 1σ errors

Fourier fit to the 71 points ($m \le 3$, $R^2 = 0.915$): $A = 0.9965 + 0.0296\cos\ell + 0.0434\sin\ell + 0.0636\cos 2\ell + 0.0366\sin 2\ell - 0.0299\cos 3\ell - 0.0114\sin 3\ell$ (ordinary least squares; reproduced exactly). Amplitudes $A_m = \sqrt{a_m^2 + b_m^2}$ with $\sigma^2_{A_m} = g^T\,\mathrm{Cov}(a_m, b_m)\,g$, $g = (a_m, b_m)/A_m$. The coefficient covariance is computed three ways:

(i) *naive*: 71 independent points, $\sigma_i = (\sigma_{\rm lo} + \sigma_{\rm hi})/2$, $\mathrm{Cov} = (X^T W X)^{-1}$ (the weights used in the earlier sheets);

(ii) $N_{\rm eff} = 16$: (i)$\times\sqrt{71/16}$ (rule of thumb: ∼16 independent sectors);

(iii) *sector-correlated* (adopted): $\mathrm{Cov} = B\,C\,B^T$, $B = (X^T X)^{-1} X^T$, $C_{ij} = \sigma_i\sigma_j \max(0,\, 1 - |\ell_i - \ell_j|/90°)$, the overlap fraction of two 90° sectors when clusters are uniform on the sky.

Model (iii) is supported by the residuals: their rms about the $m \le 3$ fit is 0.021, versus a mean error bar of 0.097 (naive $\chi^2/\nu = 0.05$). That is impossible for independent points, and it is what boxcar-smoothed noise predicts. The smoothing damps high m strongly, so (iii) gives a large error on A_1 and a small one on A_3.

	A_0	A_1 (peak 55.7°)	A_2 (peak 14.96°)	A_3 (peak 67.0°)
value	0.9965	0.0525	0.0734	0.0320
σ (iii) sector-correlated	0.0489	0.0602	0.0439	0.0212
σ (ii) $N_{\rm eff} = 16$	0.0242	0.0329	0.0336	0.0331
σ (i) naive	0.0115	0.0156	0.0160	0.0157
signal/$\sigma_{\rm (iii)}$	–	0.87	1.67	1.51

Under (iii), the amplitude estimates are mildly correlated ($|\rho| \le 0.26$); the χ^2 below ignores that. Even the octupole itself is only a 1.5σ feature.

Step 3. Set amplitudes equal, then count equations and unknowns

With $P \equiv NK$ and the axis fixed at the quadrupole peak, $\ell_0 = \ell_2 = 14.96°$:

$$\text{(E0) } N(1 + Kq_0) = A_0, \qquad \text{(E1) } P\varepsilon|o_1| = A_1, \qquad \text{(E2) } P\,q_2 = A_2, \qquad \text{(E3) } P\varepsilon|o_3| = A_3.$$

Free ε: unknowns $N, K, b_0, \varepsilon \Rightarrow$ 4 equations, 4 unknowns: an exact solve with nothing left over to test.
Fixed D/r: ε is given, so there are 3 unknowns N, K, b_0 but still 4 equations: overdetermined by one. N appears only in (E0), which is therefore satisfied exactly by $N = A_0 - Pq_0$. That leaves (E1)–(E3), three equations in (P, b_0), so one degree of freedom is left to test.

Step 4. Which ratio pins ε
Divide the equations so that P cancels:

$$\frac{A_3}{A_1} = \frac{|o_3|}{|o_1|} \;\; (\varepsilon \text{ cancels: fixes } b_0 \text{ only}), \qquad \frac{A_1}{A_2} = \varepsilon\,\frac{|o_1|}{q_2}, \qquad \frac{A_3}{A_2} = \varepsilon\,\frac{|o_3|}{q_2} = \varepsilon\,\frac{0.22806a^3}{0.49052a^2} = 0.46494\,a\,\varepsilon.$$

The dipole and octupole *both* carry ε, so A_3/A_1 says nothing about ε. The octupole-to-quadrupole ratio does, and it has a very simple form. Since $a = \cos b_0 \le 1$:

$$\varepsilon = \frac{A_3/A_2}{0.46494\,a} \;\ge\; \frac{0.4361}{0.46494} = 0.9379 \qquad \Longrightarrow \qquad D/r \le 1.066 \quad \text{(central values)}.$$

Read the other way round: if (E2) is matched, the octupole the model can produce is capped at $A_3^{\rm pred} \le 0.46494\,\varepsilon\,A_2$. That cap is $0.0171, 0.0114, 0.0085$ for $D/r = 2, 3, 4$, against the data value 0.0320. *Fixing ε smaller than* 0.938 *therefore forces the predicted octupole below the data, whatever b_0 is.*

Free-ε solve (check). $A_3/A_1 = 0.6091 = |o_3/o_1|$ gives $b_0 = 19.10°$ ($a = 0.94496$; on-axis dipole) or $31.66°$ (flipped dipole). Then $P = A_2/q_2 = 0.07338/0.43801 = 0.16753$ and $\varepsilon = A_3/(P|o_3|) = 0.99257$ ($D/r = 1.0075$), $N = A_0 - Pq_0 = 1.02260$, $K = P/N = 0.16383$. The flipped root gives $\varepsilon = 1.1019$ ($D/r = 0.9075$), $K = 0.2055$. Both reproduce the old v2 sheet.

Step 5. The least-squares objective at fixed ε

$$\chi^2(P, b_0; \varepsilon) = \sum_{m=1}^{3} \frac{\big(A_m - P\,c_m(b_0, \varepsilon)\big)^2}{\sigma_m^2}, \qquad c_1 = \varepsilon|o_1|,\; c_2 = q_2,\; c_3 = \varepsilon|o_3|, \qquad P(b_0) = \frac{\sum_m A_m c_m/\sigma_m^2}{\sum_m c_m^2/\sigma_m^2},$$

and $\chi^2(b_0)$ is minimised by a 0.05° scan plus a bounded 1-D refinement. Two recipes are used: **(LSQ)** weights $1/\sigma_m^2$ from (i). These are the weights of the earlier term-matching run, and they are nearly equal, so the absolute scale does not matter. **(Exact)** solve (E0)–(E2) exactly and treat (E3) as a pure prediction. From $A_1/A_2 = \varepsilon|o_1|/q_2$ we get b_0, then $P = A_2/q_2$, and $A_3^{\rm pred} = 0.46494\,a\,\varepsilon A_2$. Because the dipole and quadrupole are then matched exactly, every bit of misfit goes into A_3. In both recipes the misses are quoted as pulls $(A_m^{\rm pred} - A_m)/\sigma_m$ under each error model. The phases are Chris's transforms: $\phi_1 = \ell_1^{\rm data} - \ell_{\rm nat}^{(1)}$ (rotation), $\phi_2 = 0$ (the 2ℓ phase is set by ℓ_0), and the octupole stays at its natural peak $\ell_0 + 60°$, i.e. $\delta_3 = -8.00°$ from the data. R^2 is that of the transformed curve against the 71 points.

Step 6. Solve at $D/r = 2, 3, 4$: predicted harmonics vs data

				predicted amplitude			pull = (pred − data)/σ					
case	b_0	N	K	A_1	A_2	A_3	A_1, A_2 (iii)	A_3 (iii)	A_3 (ii)	A_3 (i)	$\chi^2_{\rm (iii)}$	R^2
data $\pm\sigma_{\rm (iii)}$				0.0525 ±0.060	0.0734 ±0.044	0.0320 ±0.021						0.915[a]
free ε, $D/r = 1.0075$	19.1°	1.0226	0.1638	0.0525	0.0734	0.0320	+0.00, +0.00	+0.00	+0.00	+0.00	0.00	0.897
$D/r = 2$ LSQ	6.3°	1.0314	0.1529	0.0530	0.0764	0.0177	+0.01, +0.07	−0.68	−0.43	−0.91	0.46	0.884
$D/r = 3$ LSQ	40.0°	0.9820	0.2708	0.0519	0.0766	0.0091	−0.01, +0.07	−1.08	−0.69	−1.46	1.17	0.858
$D/r = 4$ LSQ	43.3°	0.9688	0.3025	0.0519	0.0761	0.0064	−0.01, +0.06	−1.21	−0.77	−1.63	1.46	0.846
$D/r = 2$ exact, on-axis	4.1°	1.0305	0.1459	0.0525	0.0734	0.0170	−0.00, +0.00	−0.71	−0.45	−0.95	0.50	0.884
$D/r = 2$ exact, flipped	36.7°	0.9928	0.2342	0.0525	0.0734	0.0137	+0.00, +0.00	−0.86	−0.55	−1.17	0.75	0.875
$D/r = 3$ exact, flipped	40.5°	0.9806	0.2642	0.0525	0.0734	0.0086	+0.00, +0.00	−1.10	−0.71	−1.49	1.22	0.856
$D/r = 4$ exact, flipped	43.9°	0.9671	0.2982	0.0525	0.0734	0.0061	+0.00, +0.00	−1.22	−0.78	−1.65	1.49	0.846
$D/r = 3$ LSQ, b_0=0 branch	0.0°	1.0352	0.1628	0.0404	0.0826	0.0128	−0.20, +0.21	−0.91	−0.58	−1.22	0.91	0.848
$D/r = 4$ LSQ, b_0=0 branch	0.0°	1.0360	0.1659	0.0309	0.0843	0.0098	−0.36, +0.25	−1.05	−0.67	−1.41	1.29	0.801

[a] Fourier fit, octupole at its own phase. "on-axis"/"flipped" = sign of o_1 (dipole natural peak at ℓ_0 or $\ell_0 + 180°$). At $D/r \ge 2.05$ the on-axis branch can no longer reach A_1/A_2 (its maximum of $\varepsilon|o_1|/q_2$ is $1.4665\,\varepsilon$ at $b_0 = 0$), so the exact recipe has only the flipped root there.

What the table says. In every fixed-D/r row the dipole and quadrupole are reproduced to within 0.1σ (LSQ) or exactly (exact recipe); the whole cost sits in the octupole. The predicted A_3 falls roughly as ε: $0.0177, 0.0091, 0.0064$ at $D/r = 2, 3, 4$, against 0.0320. That is a miss of $0.7, 1.1, 1.2\,\sigma$ with the adopted errors ((ii): $0.4, 0.7, 0.8\,\sigma$; naive (i): $0.9, 1.5, 1.6\,\sigma$). R^2 drops from 0.897 to $0.884, 0.858, 0.846$. The curves themselves differ by little more than a 3ℓ ripple of

amplitude $\lesssim 0.032$ (Fig. 1c,d). This is the precise meaning of "the octupole misses a little": a $\sim$1–1.6σ shortfall in one harmonic that is itself only a 1.5σ detection.

How much can the octupole constrain D/r? The profile $\chi^2_{(\mathrm{iii})}(D/r)$, minimised over (P, b_0) with weights (iii) (one degree of freedom), is $0.42, 0.89, 1.28, 1.75$ at $D/r = 2, 3, 4, 6$, with minimum ≈ 0 at $D/r \approx 1$. $\Delta\chi^2 = 1$ is crossed at $D/r \approx 3.28$ and $\Delta\chi^2 = 4$ is never reached: as $\varepsilon \to 0$, $\chi^2 \to (A_3/\sigma_3)^2 = 2.28$. With the naive errors (i) the corresponding numbers are $\Delta\chi^2 = 1$ at $D/r \approx 2.16$ and an asymptote of $(A_3/\sigma_3)^2 = 4.16$. So the central values prefer $D/r \approx 1$, but the octupole alone cannot exclude $D/r = 2$–4 at even 2σ under any of the three error models. With the (iii) weights the LSQ optimum at $D/r = 3$ and 4 switches to the $b_0 = 0$ branch ($\chi^2_{(\mathrm{iii})} = 0.89, 1.28$), so the b_0 reported at those D/r is weight-dependent. The A_3 shortfall is not.

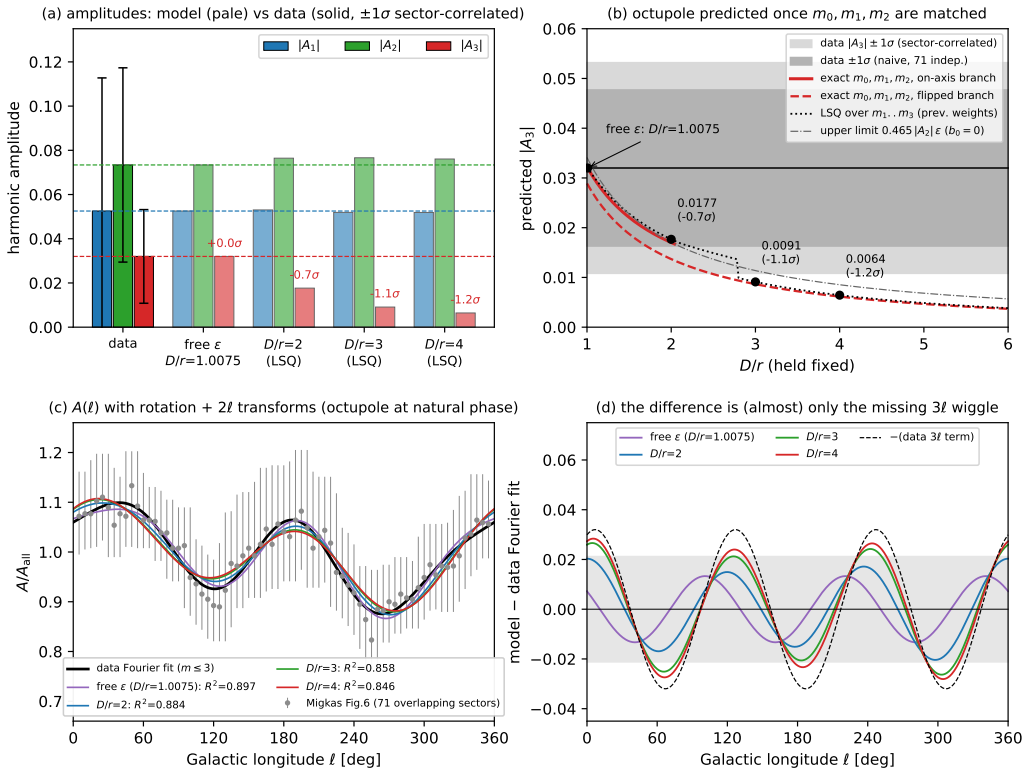


Figure 1: (a) Harmonic amplitudes: data with $\pm1\sigma$ (iii) error bars, and the model at free ε and at $D/r = 2, 3, 4$ (LSQ). The octupole pull is written above each red bar. (b) Predicted octupole vs the fixed D/r once m_0–m_2 are fitted: exact recipe (solid: on-axis branch, which exists up to $D/r = 2.05$; dashed: flipped branch), LSQ (dotted; the step at $D/r \approx 2.8$ is the switch from the on-axis to the flipped minimum), and the cap $0.465\,\varepsilon A_2$. The curve meets the data at $D/r = 1.0075$, the free-ε solution. Grey bands: data $\pm1\sigma$ (light: (iii); dark: naive (i)). (c) $A(\ell)$: the 71 digitised points, the Fourier fit, and the transformed model curves. (d) Model minus Fourier fit: at fixed D/r the residual is almost a pure 3ℓ wave, close to minus the data's octupole term. The free-ε residual is the $\delta_3 = -8°$ octupole phase offset. Shaded: $\pm\sigma_3$ (iii).

Step 7. Back to Chris's variables (LSQ solutions of Step 6; $r = 14.3$ Gpc, $\alpha = 1$)

	free ε	$D/r = 2$	$D/r = 3$	$D/r = 4$
D/r	1.0075	2.0000	3.0000	4.0000
$D = 14.3\,\mathrm{Gpc}\times D/r$: Gpc (Gly)	14.41 (47.0)	28.60 (93.3)	42.90 (139.9)	57.20 (186.6)
$K = \alpha\kappa$; $M(<D)/M_{\mathrm{crit}} = 2K/\alpha$	0.1638; 0.328	0.1529; 0.306	0.2708; 0.542	0.3025; 0.605
N	1.0226	1.0314	0.9820	0.9688
tilt b_0 (dipole branch)	19.10° (on-axis)	6.26° (on-axis)	39.96° (flipped)	43.32° (flipped)
ℓ_0	14.96°	14.96°	14.96°	14.96°
rotation ϕ_1	+40.75°	+40.75°	−139.25°	−139.25°
Chris's φ (model→data)	75.0 → 34.30°	75.0 → 34.30°	−105.0 → 34.30°	−105.0 → 34.30°
2ℓ phase ϕ_2; φ_2	0°; 60.08°	0°; 60.08°	0°; 60.08°	0°; 60.08°
Amp_1, Amp_2	0.0525, 0.0734	0.0530, 0.0764	0.0519, 0.0766	0.0519, 0.0761
Amp_3 (predicted) vs 0.0320	0.0320 (+0.0σ)	0.0177 (−0.7σ)	0.0091 (−1.1σ)	0.0064 (−1.2σ)
octupole phase offset δ_3	−8.0°	−8.0°	−8.0°	−8.0°

φ and φ_2 are identical in all columns. The transforms put the dipole and quadrupole peaks on the data by construction, so only the amplitudes and b_0 carry information about D/r. L and M/M_0 are not separately determined by this analytic form; only $K \propto M(<D)/D^3$ enters.

Caveats. (1) Chris's rotation $\phi_1 \approx 41^\circ$ and the choice $\ell_0 = \ell_2$ are extra phase freedom that the axisymmetric physics does not predict; the test here is on amplitudes only. (2) The analytic form is an $O(\varepsilon)$ tidal expansion, and the preferred solution sits at $\varepsilon \approx 1$, where that expansion is not reliable. The bound $\varepsilon \geq 0.938$ is a statement about this truncated form. (3) Error model (iii) assumes uniform clusters and purely statistical, sector-overlap-correlated error bars read from a digitised figure; the true σ_m could differ by tens of percent. Model (ii) gives σ_3 about $1.6\times$ larger, and model (i) about $1.35\times$ smaller. (4) The sector coefficients q, o assume a uniform sky outside $|b| \leq 20^\circ$; the real cluster distribution will shift them, and hence the 0.465 in the cap. (5) All earlier fixed-D/r numbers (termmatch_Dr234: N, K, b_0, R^2, analytic octupole $0.0177,\ 0.0091,\ 0.0064$) are reproduced to machine precision. The previously quoted pulls used the naive errors (i).